

LoanIQ Advanced Presentation Script (High-Scoring Version)

Preethi – Introduction

Good morning respected faculty and everyone present here. Today, we are presenting our project titled **LoanIQ – Credit-Wise Intelligent Loan Approval Prediction System**. Our system uses Machine Learning to predict whether a loan should be approved or rejected based on applicant data.

Problem Statement

In traditional banking systems, loan approval is a manual process. It is slow, inconsistent, and often biased due to human judgment. With thousands of applications being processed, even small mistakes can lead to major financial losses or rejection of deserving candidates.

(Pause) So the key question we addressed is: Can we build a system that is fast, consistent, and data-driven?

Why We Chose This Project

We chose this project because it solves a real-world financial problem, applies core machine learning concepts, and allows full pipeline implementation—from data preprocessing to deployment. Additionally, it aligns with current fintech trends where AI is increasingly used for decision-making.

Transition

Now I will explain how we handled the data and built our system. (Switch to Jupyter Notebook)

Dataset & Missing Values

We started with 1000 records and 20 features. When we analyzed the dataset, we observed missing values in every column.

Why is this important? Because machine learning models cannot process null values directly. So we handled them using mean imputation for numerical data and mode for categorical data to preserve distribution.

Akshaya – EDA

To understand the dataset better, we performed Exploratory Data Analysis.

Class Distribution

We used a pie chart to check whether our dataset is balanced.

Why? Because imbalanced data can bias the model towards one class.

Histograms

We used histograms to analyze income distribution.

We observed right-skewed data, which indicates presence of outliers.

Hence, we applied log transformation later.

Boxplot

We used boxplots to detect outliers.

Why? Outliers can mislead the model and reduce accuracy.

Correlation Heatmap

We used a correlation heatmap to understand relationships between variables.

We found Credit Score positively correlated and DTI negatively correlated with loan approval.

Insight: This helped us decide which features are important.

Preethi – Feature Engineering

Since most correlations were weak, we introduced new features.

DTI² – captures non-linear risk increase at high debt levels.
Credit Score² – strengthens effect of high credit scores.
Log Income – normalizes skewed distribution.

Result: These improved model performance by ~3–4%.

Models Used

We trained three models: Logistic Regression, KNN, and Naïve Bayes.

Why Logistic Regression?

Because it is interpretable, stable, and widely used in financial systems where explainability is important.

Evaluation Metrics

Accuracy alone is not enough in real-world systems.

Precision – avoids giving loans to risky applicants.
Recall – ensures deserving applicants are not rejected.
F1 Score – balances both.

Key Insight: In banking, both types of errors are costly.

Results

Logistic Regression achieved the best performance with 87% accuracy and balanced precision and recall.

Confusion Matrix Insight

We observed equal false positives and false negatives, indicating balanced model behavior.

Real-Life Testing

We tested our model using real inputs like our parents' financial details. The system successfully predicted their loan approval probability, showing practical applicability.

Sanjay – Demo & Future

We deployed the model using Streamlit, allowing users to input details and get instant predictions.

Future Scope

We plan to enhance this project by using advanced models like Random Forest and XGBoost, integrating real banking APIs, adding explainable AI like SHAP, and scaling it into a full fintech system.

How We Will Achieve This

We will use real datasets, optimize hyperparameters, deploy scalable backend systems, and continuously monitor model performance.

Conclusion

To conclude, our project demonstrates how machine learning can transform traditional loan approval into a fast, unbiased, and intelligent system.

Closing Line

This is not just a model, but a step toward intelligent financial decision systems. Thank you.